Calcurion Maths Worksheets

A-Level Mathematics

Topic: Mechanics – Friction

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. A particle of mass 5 kg rests on a rough horizontal surface with $\mu = 0.4$. Find the frictional force when a horizontal force of 10 N is applied. **[3]**

2. A particle of mass 3 kg rests on a rough horizontal surface with $\mu = 0.3$. Find the least horizontal force needed to move it. **[3]**

3. A particle of mass 4 kg is on a rough horizontal surface ($\mu = 0.25$). A horizontal force of 15 N is applied. Find the acceleration. **[4]**

4. A particle of mass 2 kg rests on a rough plane inclined at 20°. The coefficient of friction is 0.3. Does it remain at rest? **[4]**

5. A particle of mass 5 kg is on a rough plane inclined at 30° with $\mu = 0.4$. Find the acceleration down the plane. **[5]**

6. A particle of mass 3 kg is on a rough plane inclined at 25° with $\mu = 0.5$. Find the least force parallel to the plane needed to prevent sliding down. **[5]**

7. A particle of mass 6 kg is pulled along a rough horizontal surface ($\mu = 0.2$) by a horizontal force of 20 N. Find the acceleration. **[4]**

8. A particle of mass 4 kg is on a rough horizontal surface. The least force needed to move it is 12 N. Find μ. **[3]**

9. A particle rests on a rough plane inclined at α where $\tan\alpha = \mu$. Show it is on the point of slipping. **[3]**

10. A particle of mass 2 kg is projected up a rough plane inclined at 30° with $\mu = 0.2$ at $8\,\mathrm{m\,s^{-1}}$. Find the distance it travels up the plane. **[6]**

11. A particle of mass 5 kg is on a rough horizontal surface ($\mu = 0.3$). A force of 20 N acts at 30° to the horizontal. Find the acceleration. **[5]**

12. A particle of mass 3 kg rests on a rough plane inclined at 40°. $\mu = 0.6$. Find the frictional force. **[4]**

13. A body of mass 10 kg is on a rough horizontal floor ($\mu = 0.25$). Find the least force at 20° to the horizontal needed to move it. **[5]**

14. A particle slides down a rough plane inclined at 35° with acceleration $2\,\mathrm{m\,s^{-2}}$. Find μ. **[5]**

15. State Coulomb's laws of friction. **[3]**

Section B – Hard (Q16–25)

16. A particle of mass 4 kg is projected up a rough plane inclined at 30° ($\mu = 0.25$) with speed $10\,\mathrm{m\,s^{-1}}$. Find how far it goes up and the speed when it returns to the start. **[7]**

17. A particle of mass 5 kg rests on a rough horizontal surface ($\mu = 0.4$). A force P acts at 25° above the horizontal. Find the least P to move the particle. **[5]**

18. A particle of mass 3 kg is on a rough plane inclined at 20° ($\mu = 0.3$). A force of 15 N acts up the plane. Find the acceleration. **[5]**

19. A ladder of mass 15 kg rests against a smooth wall, base on rough ground ($\mu = 0.4$), at 60° to the ground. Does it slip? **[6]**

20. A particle of mass 2 kg is projected along a rough horizontal surface ($\mu = 0.2$) with speed $6\,\mathrm{m\,s^{-1}}$. Find how far it travels before stopping. **[4]**

21. A particle slides down a rough plane of inclination α with $\mu = \frac{1}{3}\tan\alpha$. Find the acceleration. **[4]**

22. A body of mass 8 kg is pulled up a rough plane inclined at 25° ($\mu = 0.3$) by a force of 60 N parallel to the plane. Find the acceleration. **[5]**

23. A particle of mass m is on a rough horizontal surface. The least force required to move it is P when applied horizontally and Q when applied at 30° to the horizontal. Find μ in terms of P and Q. **[6]**

24. A particle of mass 4 kg rests on a rough plane inclined at 30°. The least force down the plane to make it move is 5 N. Find μ. **[5]**

25. A particle is projected up a rough plane inclined at 45° with $\mu = 0.5$. Show that the retardation going up is $g\sqrt{2}$ and the acceleration coming down is $\frac{g}{\sqrt{2}}$. **[6]**

End of Worksheet • *Show all working clearly. Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless otherwise stated.*